

M003 Week 5 — English Worksheet (Intermediate)

Topic: Recreate the Statues (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 40 minutes

A statue is **stacked fill boxes**. One **fill** makes a solid box between **two corners**. Stack the boxes higher and higher to make legs, a body, and a head. Every corner is **(x, y, z)** — x **across**, y **up**, z **forward**. To put one box **on top** of another, make its **y** start where the box below stopped. The feet sit on the **ground (y = 0)**.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each box. Write a full sentence about what you will see. Do **not** run it.

```
fill GRAY from (3, 0, 3) to (5, 1, 5)
```

How wide (x) and deep (z) is this base? How tall (y) is it?


```
fill GRAY from (3, 2, 3) to (5, 5, 5)
```

This body starts at y = 2 — right on top of the base. Is it taller or shorter than the base?


```
fill WHITE from (3, 6, 3) to (5, 7, 5)
```

This box starts at $y = 6$. Is it near the bottom or the top of the statue?

2 · Spot the Bug 🐛

Each block is broken. Read what it should do, write the fixed code, then explain.

Bug A — The body should sit **on top** of the base. The base stops at $y = 1$, so the body should start at $y = 2$. Right now the body uses the **same y** as the base, so it lands inside it and nothing stacks up.

```
fill GRAY from (3, 0, 3) to (5, 1, 5)
fill GRAY from (3, 0, 3) to (5, 4, 5)
```

Hint: the **2nd number (y)** is height. Where should the body's bottom y be?

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences.)

Bug B — This should be the **head**, a box on top of the body at $y = 6$ – 7 . Right now both corners use **y = 6**, so it is a flat slab with no height.

```
fill WHITE from (3, 6, 3) to (5, 6, 5)
```

Hint: check the **2nd column (y)** in both corners.

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences.)

3 · Build It 📷

Walk around a statue in your world. Then build your own on your home spot. It is **3 fill boxes**: a base (legs), a body, and a head.

Recipe (every corner is **(x, y, z)**, feet on the ground $y = 0$): - base / legs: gray from (3, 0, 3) to (5, 1, 5) - body: gray from (3, 2, 3) to (5, 5, 5) - head: white from (3, 6, 3) to (5, 7, 5)

Send a **photo or video** of your statue. Then finish these — about 4 sentences.

First, I built the base by ...

The body sits on top because its **y** starts at ...

My statue is ... tall, from $y = 0$ up to $y = \dots$

The number I changed to stack each box higher was **y**, which means ...

4 · Explain It 🎥

Film yourself showing your statue. Try to use these words: **fill**, **box**, **x**, **y**, **z**, **stack**, **home spot**.

1. Show your home spot and stand on it. Say why you stand there every run.
2. Point at the base, the body, and the head. Read one corner of each out loud.
3. Show two boxes and say which one sits higher, and which number makes it higher.
4. Say in your own words what each of **x**, **y**, and **z** does.

Plan what you will say — about 4 sentences:

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.